Applied Econometrics: Session 4 Notes

Regression, Dummy Variables, Grouped Data, and Controls

This handout explains why linear regression is so useful in applied causal work, and why it is also easy to misuse. We connect regression to potential outcomes, conditional expectation functions, dummy variables, grouped data, and the choice of controls. The main lesson is practical: a regression coefficient is credible only when the comparison it represents is credible.

Reading and objective. These notes accompany Applied Econometrics Session 4 and are based on chapters 05, 06, and 07 of the Python Causality Handbook: *The Unreasonable Effectiveness of Linear Regression*, *Grouped and Dummy Regression*, and *Beyond Confounders*.
By the end of the session you should be able to explain what regression is comparing, derive the omitted-variable-bias formula, use dummy variables and grouped regressions correctly, and decide whether a proposed control is a confounder, a precision control, a treatment predictor, a mediator, or a collider.

The Big Picture

Regression is often introduced as a prediction tool. In this course we use it mostly as a comparison tool. A regression with a treatment dummy asks whether treated units have different outcomes from untreated units. A regression with controls asks whether treated and untreated units have different outcomes *after comparing units with the same or similar controls*. Dummy variables let those comparisons be flexible across groups. Grouped regression shows that many regressions are just weighted averages of group-level comparisons.

The danger is that the software always returns a coefficient. It does not know whether the coefficient is causal. That judgment comes from the research design.

The habit to build. Every time you see a regression coefficient, ask: "Which units are being compared? Why are those units comparable? What has been held fixed? What has been accidentally held fixed?"

Potential Outcomes Reminder

Let $D_i \in \{0, 1\}$ denote treatment. Unit i has two potential outcomes:

$$Y_i(1) \quad \text{if treated,} \qquad Y_i(0) \quad \text{if untreated.}$$

We observe only

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

The individual treatment effect is $Y_i(1) - Y_i(0)$, but for any one unit one of the two potential outcomes is missing. The average treatment effect is

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)].$$

For the treated units, the target is often

$$ATT = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1].$$

Definition 1 (Selection bias)
The raw treated-control difference can be decomposed as

$$\begin{aligned}\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] &= \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \\ &= ATT + \{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]\}.\end{aligned}$$

The second term is selection bias: the difference that would have existed even without treatment.

This formula is the reason we need a research design. Treated people often differ from untreated people before treatment. Students with more education may have higher wages not only because education raises wages, but also because they differ in ability, family background, local labor markets, or motivation.

Treatment Dummy Regression

Start with the simplest regression:

$$Y_i = \alpha + \tau D_i + u_i.$$

The fitted value for an untreated unit is $\hat{\alpha}$. The fitted value for a treated unit is $\hat{\alpha} + \hat{\tau}$. Therefore

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0,$$

where $\bar{Y}_1$ is the sample mean among treated units and $\bar{Y}_0$ is the sample mean among untreated units.

Theorem 1 (Dummy regression equals a difference in means)
In an OLS regression of Y_i on a constant and a binary variable D_i, the coefficient on D_i equals the treated sample mean minus the untreated sample mean.

Proof 1
OLS minimizes

$$\sum_{i:D_i=0}(Y_i - a)^2 + \sum_{i:D_i=1}(Y_i - a - b)^2.$$

For untreated units the fitted value is a; for treated units it is $a + b$. The least-squares fitted value within each group is the group mean, so $\hat{a} = \bar{Y}_0$ and $\hat{a} + \hat{b} = \bar{Y}_1$. Thus $\hat{b} = \bar{Y}_1 - \bar{Y}_0$.

Example 1 (Job training)
Suppose the average earnings of trained workers are 24,000 euros and the average earnings of untrained workers are 20,000 euros. The treatment-dummy regression coefficient is 4,000. This is a causal estimate only if the trained and untrained workers would have had the same average earnings without the program. If better workers selected into training, the coefficient overstates the effect.

Conditional Expectation Functions

The conditional expectation function, or CEF, is

$$m(x) = \mathbb{E}[Y_i \mid X_i = x].$$

It is the best mean-squared-error predictor of Y using X. In causal work, the CEF is useful because it describes how average outcomes vary with observables. It is not automatically causal. If X includes education, then $\mathbb{E}[wage \mid education = 16]$ is the average wage among people with 16 years of education. It is not automatically the wage a randomly chosen person would earn if forced to obtain 16 years of education.

Definition 2 (Conditional independence)
A control vector X_i solves selection on observables if

$$\{Y_i(1), Y_i(0)\} \perp D_i \mid X_i.$$

This says that after conditioning on X_i, treatment assignment is as good as random.

Under conditional independence and overlap, treatment-control comparisons within values of X are causal:

$$\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x] = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x].$$

Regression then becomes a way of averaging many conditional comparisons.

Population and Sample OLS

Population OLS chooses coefficients that minimize expected squared error:

$$\beta^{pop} = \arg\min_b \mathbb{E}[(Y_i - X_i'b)^2].$$

The first-order condition is

$$\mathbb{E}[X_i(Y_i - X_i'\beta^{pop})] = 0.$$

If $\mathbb{E}[X_i X_i']$ is invertible, then

$$\beta^{pop} = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i Y_i].$$

Sample OLS replaces expectations by averages. With matrix X and outcome vector Y,

$$\hat{\beta} = (X'X)^{-1}X'Y,$$

provided $X'X$ is invertible. The residual vector $\hat{u} = Y - X\hat{\beta}$ satisfies

$$X'\hat{u} = 0.$$

So each included regressor is uncorrelated with the OLS residual in the sample. This is a mechanical property of OLS. It is not an assumption and it is not evidence that the model is causal.

Regression Anatomy and FWL

Suppose we estimate

$$Y_i = \alpha + \tau D_i + X_i'\gamma + u_i.$$

The Frisch-Waugh-Lovell theorem tells us exactly what $\hat{\tau}$ is doing.

Theorem 2 (Frisch-Waugh-Lovell)
The coefficient $\hat{\tau}$ from the regression of Y on a constant, D, and X is equal to the coefficient from a regression of residualized Y on residualized D, where both residualizations are from regressions on the constant and X.

Let

$$\tilde{D}_i = D_i - \widehat{\mathbb{E}}[D_i \mid X_i], \qquad \tilde{Y}_i = Y_i - \widehat{\mathbb{E}}[Y_i \mid X_i].$$

Then

$$\hat{\tau} = \frac{\sum_i \tilde{D}_i \tilde{Y}_i}{\sum_i \tilde{D}_i^2}.$$

Thus multiple regression first removes from treatment the part predictable from controls. Then it asks whether the remaining treatment variation predicts the remaining outcome variation.

> **Plain-language FWL.** The coefficient on treatment comes from comparing treated and untreated observations that differ in treatment after accounting for the controls. If the controls almost perfectly predict treatment, there is very little residual treatment variation left, and the estimate becomes noisy.

When Regression Adjustment Is Causal

Regression adjustment is credible when the conditional comparisons are credible. A useful textbook condition is:

$$\{Y_i(1), Y_i(0)\} \perp D_i \mid X_i \quad \text{and} \quad 0 < P(D_i = 1 \mid X_i) < 1.$$

The first part is conditional independence. The second part is overlap: for the same values of X, we need both treated and untreated units.

Assumption 1 (No bad controls)
The controls X_i should be measured before treatment or otherwise not be caused by treatment, and they should not be colliders created by conditioning on common effects.

Assumption 2 (Functional-form adequacy)
If treatment effects or baseline outcomes vary strongly with X_i, the regression specification must be flexible enough to approximate the relevant conditional means. Linearity is convenient, not magic.

For weak students, the best way to think about this is blocked experiments. If we compare treated and untreated men with similar age, education, and region, we are pretending that treatment was randomized inside those blocks. Regression does this blocking smoothly, with algebra rather than by manually forming cells.

Omitted-Variable Bias

Suppose the true causal equation is

$$Y_i = \alpha + \tau D_i + \gamma Z_i + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid D_i, Z_i] = 0,$$

but we estimate the short regression

$$Y_i = a + bD_i + e_i.$$

Then the population short-regression coefficient is

$$b = \tau + \gamma\delta,$$

where δ is the coefficient from the population regression of Z_i on D_i:

$$Z_i = \pi + \delta D_i + v_i.$$

Theorem 3 (Omitted-variable bias formula)
With one omitted variable Z_i and one included treatment D_i,

$$\text{Bias}(b) = b - \tau = \gamma \frac{\text{Cov}(D_i, Z_i)}{\text{Var}(D_i)}.$$

Proof 2
The short-regression coefficient is

$$b = \frac{\text{Cov}(D_i, Y_i)}{\text{Var}(D_i)}.$$

Substitute the true model:

$$\begin{aligned} b &= \frac{\text{Cov}(D_i, \alpha + \tau D_i + \gamma Z_i + \varepsilon_i)}{\text{Var}(D_i)} \\ &= \tau + \gamma \frac{\text{Cov}(D_i, Z_i)}{\text{Var}(D_i)} + \frac{\text{Cov}(D_i, \varepsilon_i)}{\text{Var}(D_i)}. \end{aligned}$$

The last term is zero by the true exogeneity condition, giving the result.

> **Sign intuition.** Bias is positive if the omitted variable raises the outcome and treated units have more of it. Bias is negative if the omitted variable raises the outcome and treated units have less of it. Always write the two signs: effect of omitted variable on Y, and association of omitted variable with D.

Example 2 (Education and wages)
Let D_i be years of education and Z_i be ability. If ability raises wages ($\gamma > 0$) and high-ability people obtain more education ($\text{Cov}(D_i, Z_i) > 0$), then a wage regression omitting ability overstates the return to education. If a policy forces low-ability students to stay in school longer, the sign of the association could differ in that sample. The bias formula is about the population being studied.

Dummy Variables

A dummy variable is a variable that takes values zero or one. Dummies are used for binary treatments, categories, fixed effects, and flexible group means.

For a binary dummy G_i,

$$Y_i = \alpha + \beta G_i + u_i$$

has fitted values

$$\hat{Y}_i = \begin{cases} \hat{\alpha}, & G_i = 0, \\ \hat{\alpha} + \hat{\beta}, & G_i = 1. \end{cases}$$

So $\hat{\beta}$ is the difference between group means.

The Dummy-Variable Trap

Suppose education has three categories: high school, college, graduate. If the regression includes a constant and all three category dummies,

$$1 = 1\{HS\} + 1\{College\} + 1\{Graduate\}$$

for every observation. One column is an exact linear combination of the others. Then $X'X$ is not invertible.

The usual solution is to omit one category. That omitted category becomes the reference group:

$$Y_i = \alpha + \beta_1 1\{College_i\} + \beta_2 1\{Graduate_i\} + u_i.$$

Here α is the high-school mean, β_1 is college minus high school, and β_2 is graduate minus high school.

> **Common mistake.** Students often say the omitted group is "ignored." It is not ignored. It is the benchmark. Every included dummy coefficient is interpreted relative to it.

Saturated Dummies

A saturated dummy regression includes one dummy for every group, except the reference category if there is a constant. With only group dummies, it reproduces the group means exactly. If group g has mean $\bar{Y}_g$, the fitted value for all observations in group g is $\bar{Y}_g$.

This is why dummy regression is flexible. It does not force a straight-line relationship between a categorical variable and the outcome. If wage rises from 12 to 16 years of education and then flattens, education dummies can show that pattern. A single linear education coefficient cannot.

Grouped Regression and Weights

Sometimes data are available only as cell means. Suppose each group g has mean outcome $\bar{Y}_g$, mean regressors $\bar{X}_g$, and cell size n_g. Individual-level OLS solves

$$\min_b \sum_i (Y_i - X_i'b)^2.$$

If all regressors are constant within group, the same coefficient can be obtained from weighted grouped OLS:

$$\min_b \sum_g n_g (\bar{Y}_g - \bar{X}_g'b)^2.$$

Theorem 4 (Why grouped weights matter)
When grouped observations represent different numbers of individual observations, the group-level regression should weight each group by its cell size if the target is the individual-level regression.

Unweighted grouped regression gives each group the same importance. That may be the right estimand if each group is itself the unit of interest. It is not the same as an individual-level regression when one group contains 10 people and another contains 10,000.

Example 3 (Education cells)
Imagine wage data grouped by years of education. There may be many people with 12 years of education and few people with 20 years. If we regress cell mean wages on education without weights, the small 20-year cell receives the same weight as the large 12-year cell. Weighted grouped OLS gives each cell weight proportional to the number of people in it.

Interactions

Interactions allow the treatment effect to differ across groups. With a binary treatment D_i and a binary group G_i,

$$Y_i = \alpha + \tau D_i + \eta G_i + \theta(D_i G_i) + u_i.$$

The treatment effect for $G_i = 0$ is τ. The treatment effect for $G_i = 1$ is $\tau + \theta$. Therefore θ is a difference in treatment effects across groups.

> **Interpretation rule.** Do not interpret the coefficient on D_i alone as "the treatment effect" when the model contains $D_i G_i$. It is the treatment effect for the reference group $G_i = 0$.

Interactions are useful when theory suggests heterogeneity. A training program may affect younger workers differently from older workers. A class-size reduction may matter more in early grades. A tax change may affect liquidity constrained households more than unconstrained households.

Controls Beyond Confounders

Not every control has the same role. The key control types are:

Confounder	Causes D and Y	Usually include
Outcome predictor	Predicts Y only	Can improve precision
Treatment predictor	Predicts D only	Can hurt precision
Mediator	$D \to M \to Y$	Do not include for total effect
Collider	$D \to C \leftarrow U \to Y$	Do not include

Precision Controls

In a randomized experiment, treatment is independent of potential outcomes. Controls are not needed for identification. But controls that strongly predict the outcome can reduce residual variance and improve precision:

$$Y_i = \alpha + \tau D_i + X_i'\gamma + u_i.$$

If X_i explains a large part of Y_i but is unrelated to D_i by randomization, including X_i can make $\hat{\tau}$ more precise without changing its causal interpretation.

Treatment Predictors

A variable that predicts treatment but not the outcome can reduce useful variation in treatment. In FWL language, it makes $\tilde{D}_i$ smaller without making $\tilde{Y}_i$ much cleaner. Standard errors may rise. This is why "include everything that predicts treatment" is not a good rule.

Mediators

A mediator is caused by treatment and then affects the outcome:

$$D_i \to M_i \to Y_i.$$

If the target is the total effect of treatment, do not control for M_i. Controlling for the mediator blocks part of the treatment effect.

Example 4 (Email reminder)
Suppose an email reminder increases payment partly because it makes people log into an online account. If the outcome is payment and the mediator is login, then controlling for login estimates the effect of the email holding login fixed. That is not the total effect of the email.

Colliders

A collider is a common effect of two variables. Conditioning on it can create a false association. A typical structure is

$$D_i \to C_i \leftarrow U_i \to Y_i.$$

Even if D_i was randomized, controlling for C_i can open a path between D_i and unobserved U_i, biasing the coefficient on treatment.

> **Selection is often collider conditioning.** If we analyze only people who applied for a program, only firms that survived, or only loans that were approved, we may be conditioning on

> a variable affected by both treatment and unobserved determinants of the outcome.

Using the Python Example

The companion script

```
lectures/code/lecture-4-regression-controls.py
```

contains three classroom examples.

1. **Omitted-variable bias.** The script simulates education, wages, latent ability, and family background. The short regression is biased because education is correlated with omitted determinants of wages.
2. **Dummy and grouped regression.** The script compares a linear education regression with a saturated education-dummy regression and shows that weighted grouped OLS matches the individual-level regression when the weights are cell sizes.
3. **Bad controls.** The script shows how adding a mediator changes the estimand and how controlling for a collider can create bias even when treatment was randomized.

When using the script in class, ask students to predict the sign of each coefficient before looking at the output. Prediction forces them to reason about the comparison rather than memorize a formula.

Practical Control-Selection Rules

1. Start with the estimand. Are you trying to estimate a total effect, a direct effect, an ATT, or a descriptive association?
2. Draw a causal story before opening the data. List variables that cause treatment, cause the outcome, are caused by treatment, or are selection rules.
3. Include pre-treatment confounders when the identification strategy is selection on observables.
4. Include strong pre-treatment outcome predictors when they improve precision and do not damage overlap.
5. Be cautious with variables that mostly predict treatment. They may make standard errors larger by removing treatment variation.
6. Do not control for mediators if the target is the total effect.

7. Do not control for colliders or post-treatment selection variables.
8. Check overlap. Regression cannot credibly compare treated units in regions of X where no controls exist.
9. Show robustness, but do not treat a long list of controls as proof of causality.

Worked Problems

Problem 1: Treatment Dummy

In a sample of 100 students, 40 attend tutoring and have average exam score 78. The 60 non-tutored students have average score 70. What is the coefficient from regressing score on a constant and a tutoring dummy?

Solution. The coefficient is the treated mean minus the untreated mean:

$$\hat{\tau} = 78 - 70 = 8.$$

The intercept is 70. The fitted score is 70 for non-tutored students and 78 for tutored students. This is causal only if tutored and non-tutored students would have had the same average score without tutoring, or if assignment to tutoring was random.

Problem 2: OVB Sign

You regress wages on education but omit parental income. Parental income raises wages through networks and school quality. Students from richer families obtain more education. What is the likely sign of omitted-variable bias?

Solution. The omitted variable has a positive effect on wages: $\gamma > 0$. It is positively associated with education: $\mathrm{Cov}(D, Z) > 0$. Therefore the omitted-variable bias is positive. The short regression likely overstates the causal return to education.

Problem 3: Dummy Trap

You include a constant, a male dummy, and a female dummy in a regression. Why does the software drop one variable?

Solution. For every observation,

$$1 = Male_i + Female_i.$$

The constant is exactly the sum of the two dummies. The design matrix has perfect multicollinearity, so $X'X$ is singular. Dropping one dummy creates a reference group.

Problem 4: Grouped Weights

A grouped dataset has cell means by region. Region A has 1,000 people and region B has 20 people. You want the individual-level relationship between income and education. Should the grouped regression weight the regions equally?

Solution. No. If the target is the individual-level regression, region A should receive 1,000 units of weight and region B should receive 20. Equal weights would estimate a region-level relationship where each region counts the same.

Problem 5: Mediator

A training program increases certification, and certification raises wages. You regress wages on training and certification. What does the training coefficient estimate?

Solution. It estimates the effect of training holding certification fixed. It does not estimate the total effect of training, because one channel from training to wages has been blocked.

Checklist for Students

Before trusting a regression coefficient, check the following.

1. Can I state the estimand in words?
2. Can I write the regression equation and interpret every coefficient?
3. For a treatment coefficient, do I know which units are being compared?
4. Are the controls pre-treatment?
5. Are any controls mediators or colliders?
6. Is there overlap between treated and untreated units after controls?
7. If dummies are used, do I know the reference group?
8. If interactions are used, do I know which group the main effect refers to?
9. If grouped data are used, are the weights aligned with the estimand?
10. Can I explain the likely direction of omitted-variable bias?

Practice Questions

1. Prove that the coefficient on a binary treatment dummy equals the difference in sample means.
2. Derive the omitted-variable-bias formula for one omitted variable.
3. Give an example of a variable that is a confounder in one study but a mediator in another.
4. In the regression

$$Y_i = \alpha + \tau D_i + \eta Female_i + \theta(D_i \times Female_i) + u_i,$$

   what is the treatment effect for men? What is the treatment effect for women?
5. Explain why a saturated dummy regression produces group means.
6. A researcher has grouped data with cell sizes. When should she use weighted grouped OLS? When might unweighted grouped OLS be the right object?
7. Why can an outcome predictor improve precision in a randomized experiment?
8. Why can a treatment predictor increase the standard error of the treatment coefficient?
9. Draw a simple DAG with a collider and explain why conditioning on the collider creates bias.
10. Run `lectures/code/lecture-4-regression-controls.py`. For each example, write one paragraph explaining what changed across specifications and why.